

Recursion

Q1) Write a program to display the message “Hello World” 5 times using recursion.

Q2) Write a program to print numeric counting from 1 to 10 using recursion.

1
2
3
4
5
6
7
8
9
10

Q3) Write a program to print multiplication table of any number using recursion.
Table number & length should be taken as an input from user.

Enter a number to enter its multiplication table: 2

Enter length of multiplication table: 13

Multiplication Table of 2

Length of 13

2 X 1 = 2
2 X 2 = 4
2 X 3 = 6
2 X 4 = 8
2 X 5 = 10
2 X 6 = 12
2 X 7 = 14
2 X 8 = 16
2 X 9 = 18
2 X 10 = 20
2 X 11 = 22
2 X 12 = 24
2 X 13 = 26

Q4) Generate the following series. Use recursion for this purpose. See example output.

- a. Counting: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
- b. Reverse counting: 10, 9, 8, 7, 6, 5, 4, 3, 2, 1
- c. Even: 0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20
- d. Odd: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19
- e. Series: 2k, 4k, 6k, 8k, 10k, 12k, 14k, 16k, 18k, 20k

Q5) Write a program to print multiples of 5 ranging 1 to 100 using recursion
5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100

Q6) Write a program to repeatedly print the value of the variable **num** which is input by user. Value should be decreasing by 0.5 each time, as long as x Value remains positive. Use recursion for this question.

Enter a number: 5

5, 4.5, 4, 3.5, 3, 2.5, 2, 1.5, 1, 0.5, 0

Q7) The even/odd reporter

Write a function that uses recursion to check whether the current number is even or odd, and report that to the screen (e.g. "2 is even"). The function does it from 0 to 20.

0 is Even
1 is Odd
2 is Even
3 is Odd
4 is Even
5 is Odd
6 is Even
7 is Odd
8 is Even
9 is Odd
10 is Even
11 is Odd
12 is Even
13 is Odd
14 is Even
15 is Odd
16 is Even
17 is Odd
18 is Even
19 is Odd
20 is Even

Q8) Write a program to calculate the product of the odd integers from 1 to 7 using recursion. The product of the odd integers from 1 to 7 is 105

Q9) Write a program that will write out a wedge of stars using recursion. The user will enter the initial number of stars, and the program will write out lines of stars where each line has one few star than the previous line.

Initial number of stars: 7

```
*****  
*****  
*****  
****  
***  
**  
*
```

Q10) Write a program to create the following patterns using recursion. Take number of lines as an input.

a.

```
*****  
*****  
*****  
*****
```

b.

```
*  
**  
***  
****  
*****
```

c.

```
*****  
***  
**  
*
```